

Title: Understanding APIs — A Beginner's Guide

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1. Introduction

This document explains what an API is for beginners with no prior technical background.

2. The Broker Analogy.

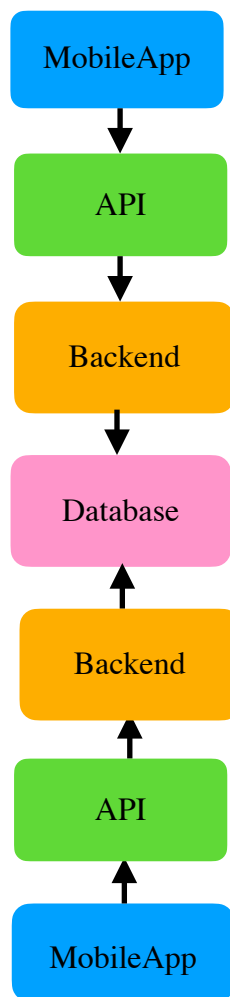
Let's look at a quick analogy. Imagine you are new to a city and searching for an apartment. Meanwhile, a property owner in the same neighbourhood is looking for a tenant. To find each other, you both use a real estate broker. The broker acts as a middleman, facilitating seamless communication and ensuring both sides follow the correct procedures.

In the digital world, the API (Application Programming Interface) is that broker. The house hunter and the property owner are two different software applications that need to communicate to get a specific job done. The API acts as the middleman that understands both parties, enforces the rules of the interaction, and fulfils their needs.

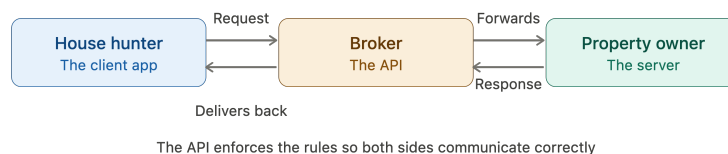
3. What is an API? (Formal Definition)

API stands for Application Programming Interface. It is a set of rules, protocols, and definitions that allows two applications to communicate with each other, regardless of how complex they are.

The client sends a request through the API. The server processes the request and returns a response, which the API delivers to the client.



4. How It Works — Diagram



5. Real-World Examples

5.1 Weather App

We all have a weather app on our phones. These apps do not collect weather data themselves. Instead, they get that information by submitting requests to weather APIs. These APIs retrieve real-time data from weather database servers and pass it back to the app, which then displays it to us.

5.2 Payment API

While using E-commerce websites, we pay using Payment APIs which ensures a secure transaction session without leaving the E-commerce site. APIs being middle man introduces different layers of security between client and server.

5.3 Google Login

Third-party apps use Google login to authenticate users. This happens when the third-party app sends a request to Google APIs, which fetch basic information about the user. This allows them to log into third-party apps without the need to generate a new password.

6. Conclusion

Do you think API is an app? No its not an app, it is only a set of rules and protocols and definitions for the effective communication between two applications. APIs quietly power many digital services we use every day. From weather apps to online payments, they help applications communicate quickly and securely behind the scenes. Understanding APIs is an important first step toward understanding how modern software systems work together.

7. Glossary

API (Application Programming Interface)

A set of rules and protocols that allows two software applications to communicate with each other.

Client

The application that sends a request — in our analogy, the house hunter.

Server

The application that receives the request and sends back a response — in our analogy, the property owner.

Request

A message sent by the client asking the server for information or action.

Response

The information or result the server sends back after processing a request.